Introduction to Business Intelligence & Power BI

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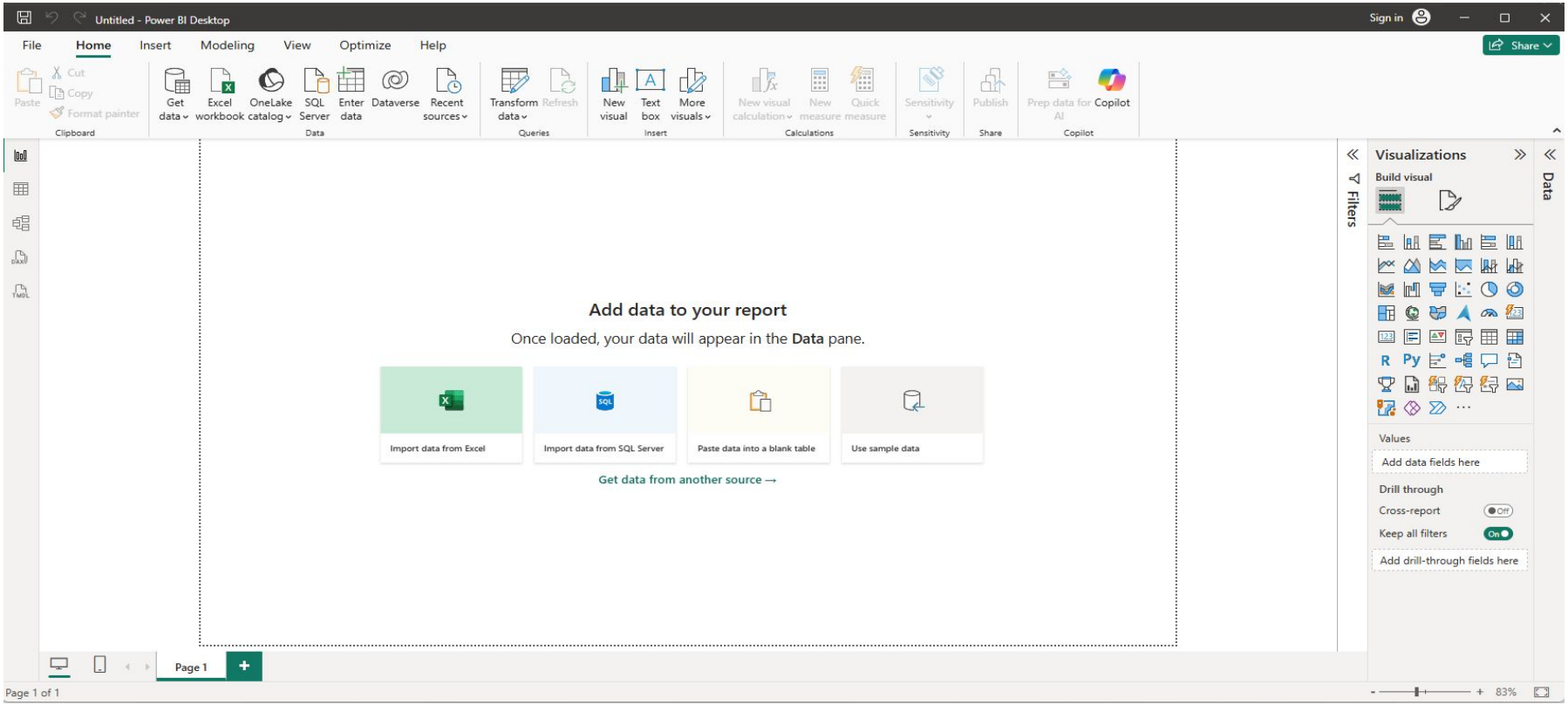
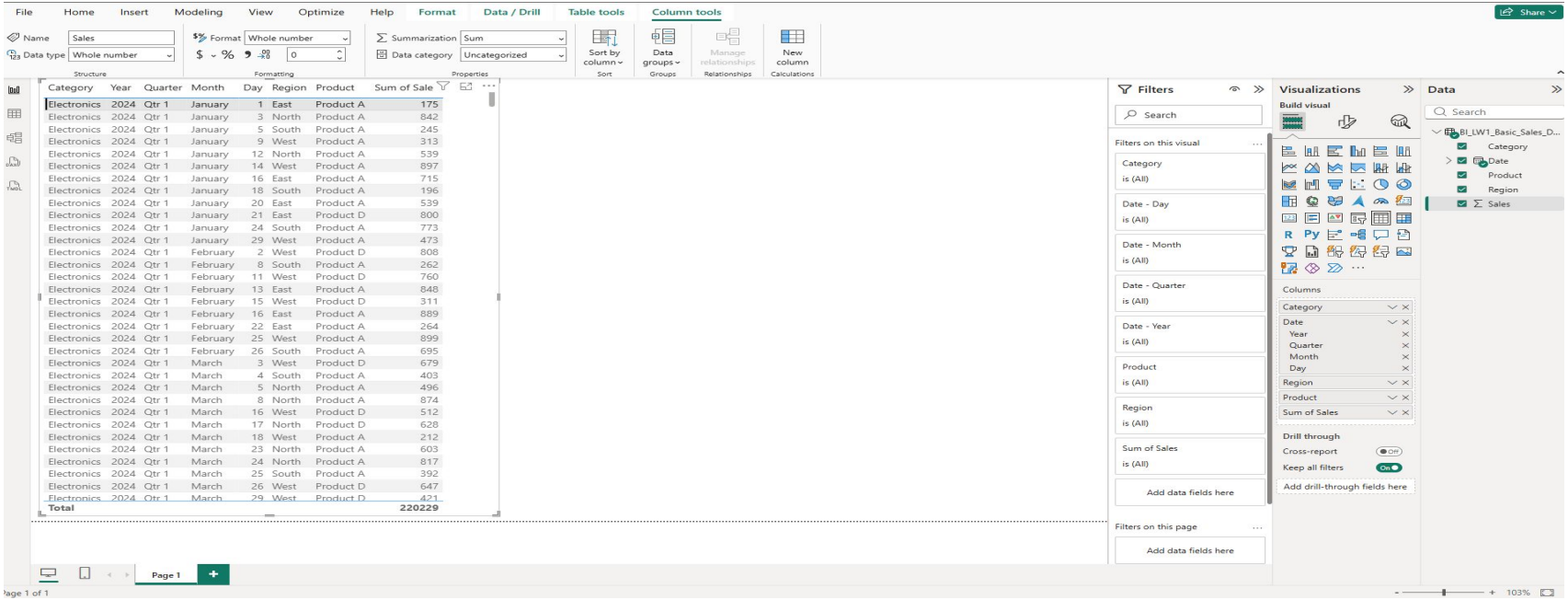
IS 107 - Business Intelligence

Part 1 | Step 1: Open Power BI Desktop

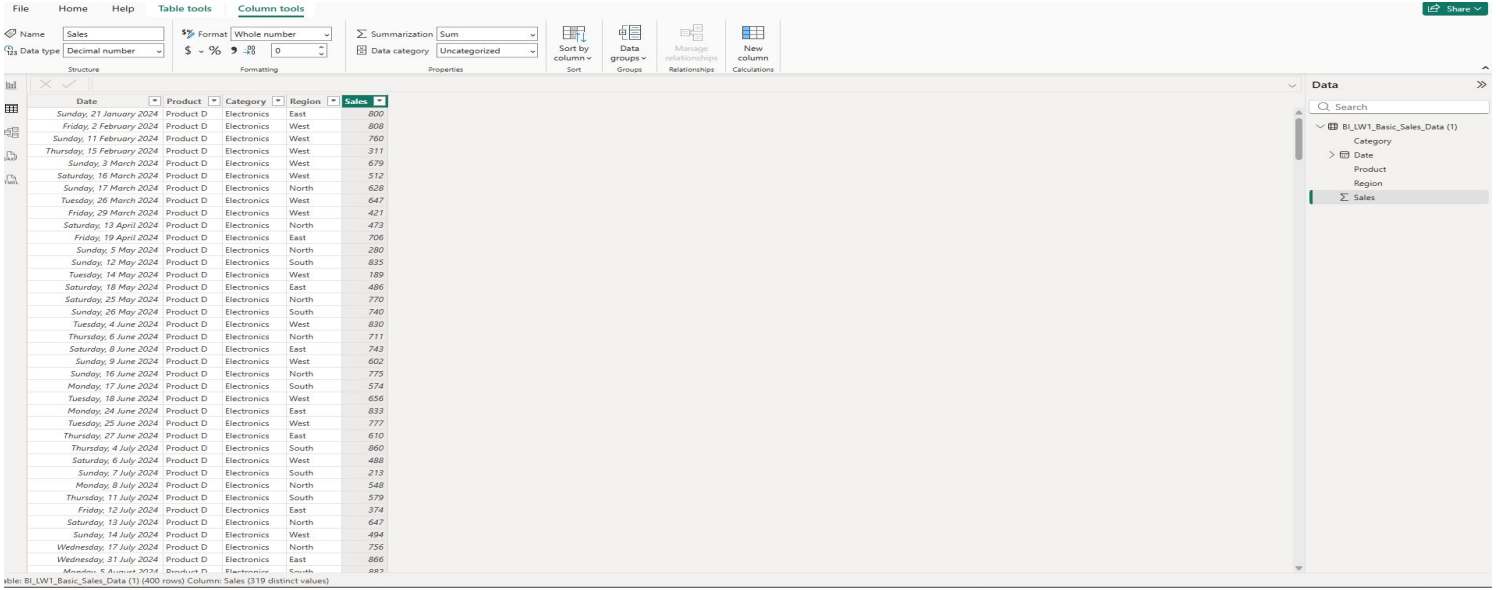


Step 2: Load the Dataset



Category	Year	Quarter	Month	Day	Region	Product	Sum of Sale
Electronics	2024	Qtr 1	January	1	East	Product A	175
Electronics	2024	Qtr 1	January	3	North	Product A	842
Electronics	2024	Qtr 1	January	5	South	Product A	245
Electronics	2024	Qtr 1	January	9	West	Product A	313
Electronics	2024	Qtr 1	January	12	North	Product A	539
Electronics	2024	Qtr 1	January	14	West	Product A	897
Electronics	2024	Qtr 1	January	16	East	Product A	715
Electronics	2024	Qtr 1	January	18	South	Product A	196
Electronics	2024	Qtr 1	January	20	East	Product A	539
Electronics	2024	Qtr 1	January	21	East	Product D	800
Electronics	2024	Qtr 1	January	24	South	Product A	773
Electronics	2024	Qtr 1	January	29	West	Product A	473
Electronics	2024	Qtr 1	February	2	West	Product D	808
Electronics	2024	Qtr 1	February	8	South	Product A	262
Electronics	2024	Qtr 1	February	11	West	Product D	760
Electronics	2024	Qtr 1	February	13	East	Product A	848
Electronics	2024	Qtr 1	February	15	West	Product D	311
Electronics	2024	Qtr 1	February	16	East	Product A	889
Electronics	2024	Qtr 1	February	22	East	Product A	264
Electronics	2024	Qtr 1	February	25	West	Product A	899
Electronics	2024	Qtr 1	February	26	South	Product A	695
Electronics	2024	Qtr 1	March	3	West	Product D	679
Electronics	2024	Qtr 1	March	4	South	Product A	403
Electronics	2024	Qtr 1	March	5	North	Product A	496
Electronics	2024	Qtr 1	March	8	North	Product A	874
Electronics	2024	Qtr 1	March	16	West	Product D	512
Electronics	2024	Qtr 1	March	17	North	Product D	628
Electronics	2024	Qtr 1	March	18	West	Product A	212
Electronics	2024	Qtr 1	March	23	North	Product A	603
Electronics	2024	Qtr 1	March	24	North	Product A	817
Electronics	2024	Qtr 1	March	25	South	Product A	392
Electronics	2024	Qtr 1	March	26	West	Product D	647
Electronics	2024	Qtr 1	March	29	West	Product D	421
Total							220229

Step 3: Verify Data in Data View

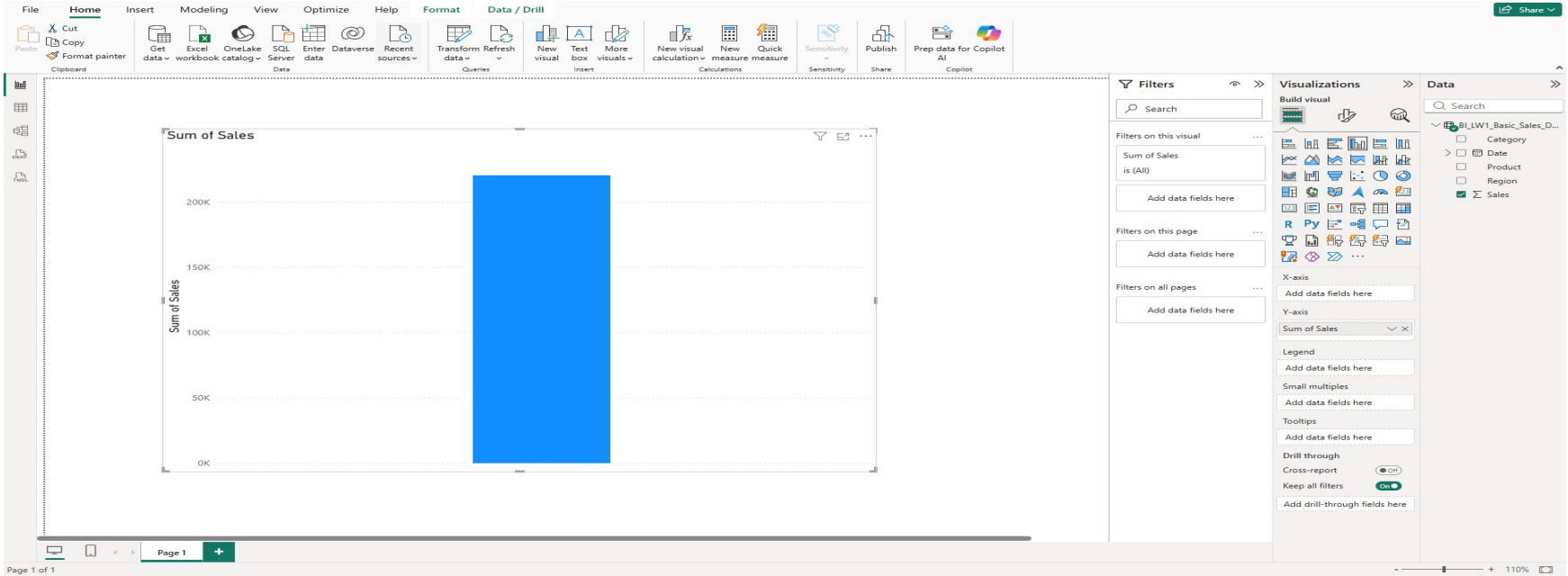


Date	Product	Category	Region	Sales
Sunday, 21 January 2024	Product D	Electronics	East	800
Friday, 2 February 2024	Product D	Electronics	West	808
Sunday, 11 February 2024	Product D	Electronics	West	760
Thursday, 15 February 2024	Product D	Electronics	West	311
Sunday, 3 March 2024	Product D	Electronics	West	679
Saturday, 16 March 2024	Product D	Electronics	West	512
Sunday, 17 March 2024	Product D	Electronics	North	628
Tuesday, 26 March 2024	Product D	Electronics	West	647
Friday, 29 March 2024	Product D	Electronics	West	421
Saturday, 13 April 2024	Product D	Electronics	North	473
Friday, 19 April 2024	Product D	Electronics	East	706
Sunday, 5 May 2024	Product D	Electronics	North	280
Sunday, 12 May 2024	Product D	Electronics	South	835
Tuesday, 14 May 2024	Product D	Electronics	West	189
Saturday, 18 May 2024	Product D	Electronics	East	486
Saturday, 25 May 2024	Product D	Electronics	North	770
Sunday, 26 May 2024	Product D	Electronics	South	740
Tuesday, 4 June 2024	Product D	Electronics	West	830
Thursday, 6 June 2024	Product D	Electronics	North	711
Saturday, 8 June 2024	Product D	Electronics	East	743
Sunday, 9 June 2024	Product D	Electronics	West	602
Sunday, 16 June 2024	Product D	Electronics	North	775
Monday, 17 June 2024	Product D	Electronics	South	574
Tuesday, 18 June 2024	Product D	Electronics	West	656
Monday, 24 June 2024	Product D	Electronics	East	833
Tuesday, 25 June 2024	Product D	Electronics	West	777
Thursday, 27 June 2024	Product D	Electronics	East	610
Thursday, 4 July 2024	Product D	Electronics	South	860
Saturday, 6 July 2024	Product D	Electronics	West	488
Sunday, 7 July 2024	Product D	Electronics	South	213
Monday, 8 July 2024	Product D	Electronics	North	548
Thursday, 11 July 2024	Product D	Electronics	South	579
Friday, 12 July 2024	Product D	Electronics	East	374
Saturday, 13 July 2024	Product D	Electronics	North	647
Sunday, 14 July 2024	Product D	Electronics	West	494
Wednesday, 17 July 2024	Product D	Electronics	North	756
Wednesday, 31 July 2024	Product D	Electronics	East	866

Question:

- Are all columns visible? Yes all columns are visible.
- Is “Date” formatted as Date? Yes the date are formatted as a date.
- Is “Sales” formatted as Decimal Number? No, the sale is formatted as whole number.

PART 3: Step 1: Quick Visualization

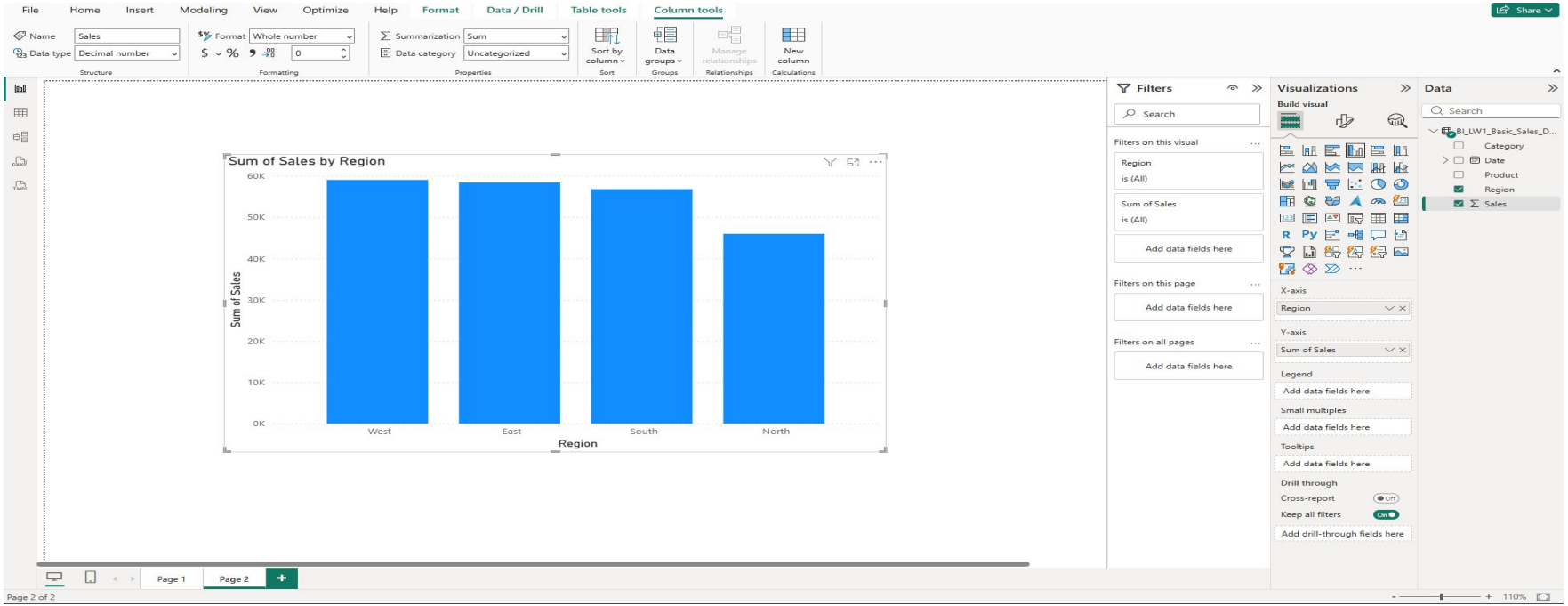


Question:

- What type of chart was created? Column Chart

- What does it show? Showing a single total sales value of approximately $220,000.

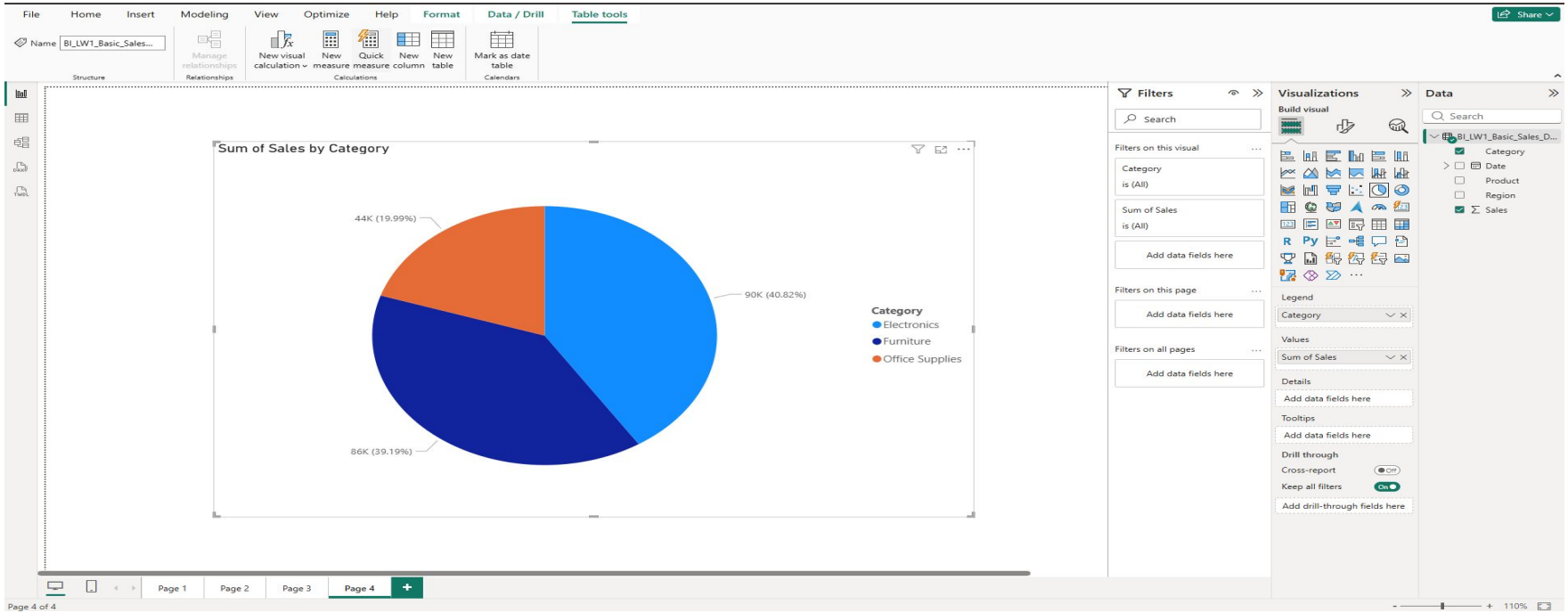
Step 2: Create a Sales by Region Chart



Question:

- Which region has highest sales? The region of the West.

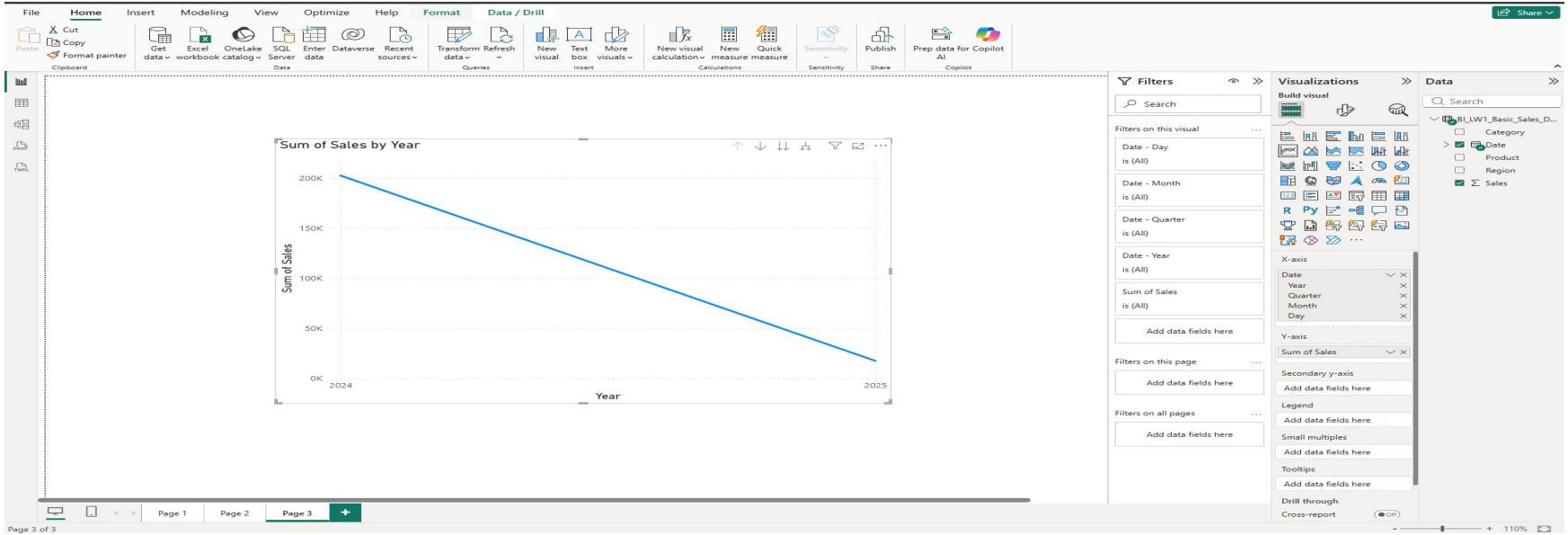
Step 3: Sales by Category



Question:

- Which category dominates? The electronics category.
- Is the distribution balanced? Yes because it is base on pie chart it depents on which is the most high sales.

Step 4: Sales Over Time



Question:

- Is there growth? no there is no growth.
- Any noticeable trend? In the 2024 the sale is 202487 and in 2025 the sale drop to 17742.

PART 4: Basic Data Insight Interpretation Question:

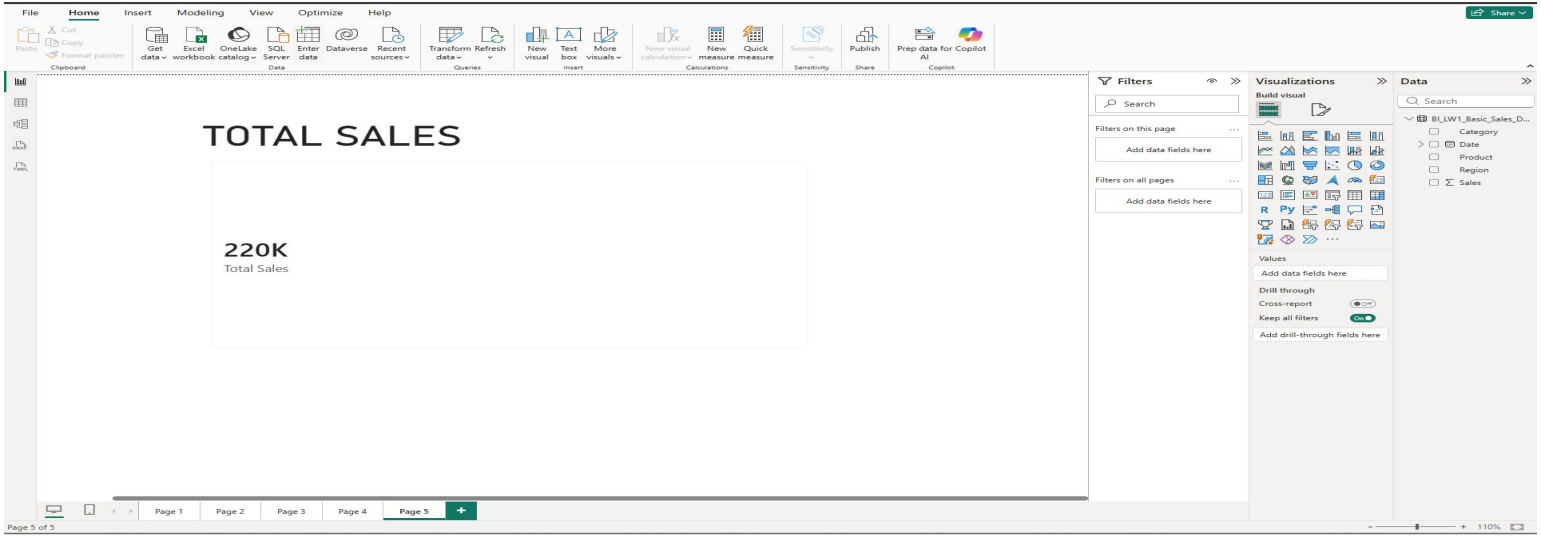
- Which region contributes most revenue? West contributes the most revenue at $59,042.60
- Which product category performs best? Electronics is the best-performing category with $89,904.20 in sales
- Are sales consistent across dates? No — sales are NOT consistent.
- What business recommendation can you suggest? Invest in Electronics product line expansion in the North region

LABORATORY QUESTIONS
Part A – Technical Questions

1. What are the five columns in the dataset? Date, Product, Category, Region, Sales
2. What data type is assigned to the “Sales” column? Decimal number (Float/Float64)
3. Which Power BI view allows you to see raw data? Data view (the table icon in the left sidebar)
4. What chart type is best for showing trends over time? Line chart
5. What aggregation is automatically applied to Sales? Sum Part B – Analytical Questions
6. Which region has the highest total sales? West with $59,042.60
7. Which category has the lowest performance? Office Supplies with $44,023.60
8. Are sales increasing, decreasing, or stable? Stable/Slightly Decreasing — Monthly sales fluctuate between ~$13,600-$21,300 with no clear upward trend (first half avg: $16,253 vs second half avg: $15,209)
9. If you were a manager, which region would you prioritize? North — It has the lowest total sales ($45,985) and fewest transactions (86), indicating untapped potential
10. Provide one actionable recommendation based on the data. Increase marketing investment in the North region and introduce promotions for Office Supplies category, which is underperforming compared to Electronics and Furniture

ENHANCEMENT SECTION

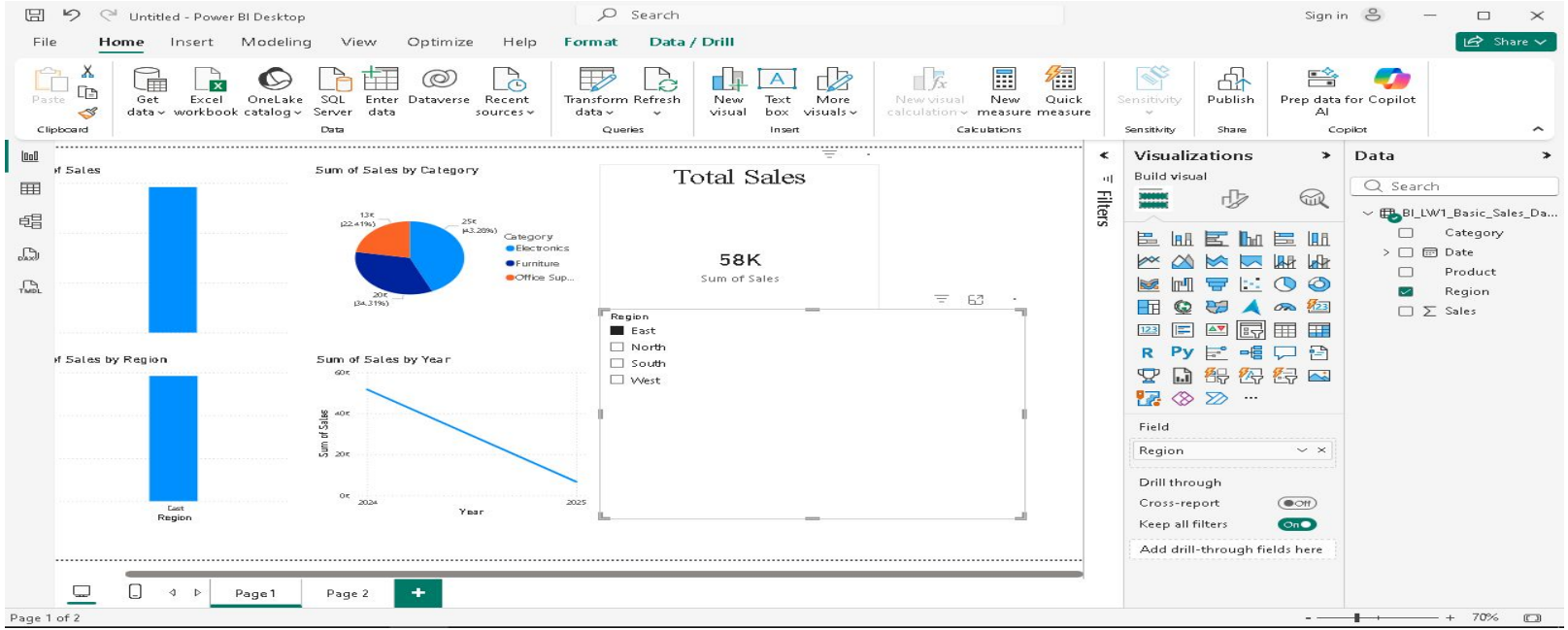
Task 1: Add a Card Visualization



Question:

- What is the total sales amount? The total sale is 220K.

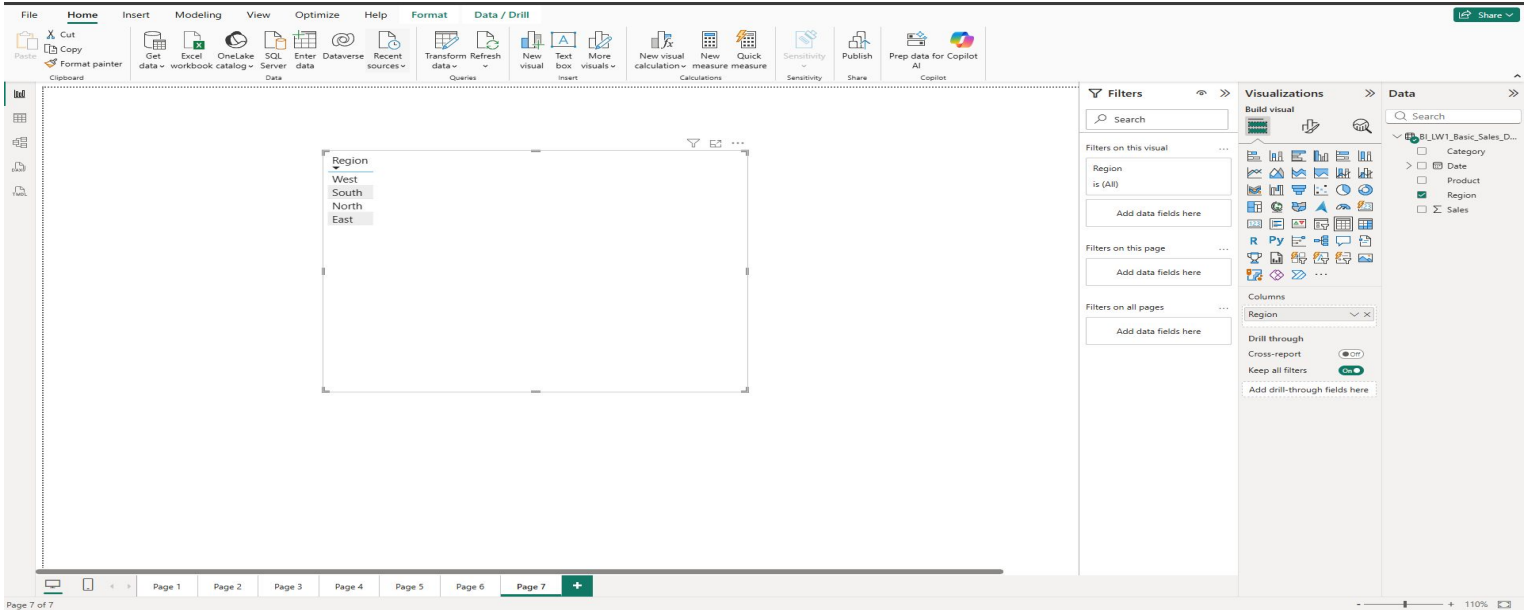
Task 2: Add Slicer



Question

- ‘What happens to other visuals when you click a region? The other visual will change.
- Why is filtering important in BI?Filtering is important in BI because it transforms overwhelming data into actionable insights

Task 3: Sort Sales



Question:

- Does sorting improve readability? Yes
- Why? because it will show which country is low to high sales.

Task 4: Identify Outliers

- Which region is significantly higher or lower? West is significantly higher ($59K) and North is significantly lower ($46K) — a $13K gap.
- What might explain that difference? Possible explanations: North may have less store presence, weaker marketing, lower customer demand, fewer product offerings, or stronger competition compared to West.